

MoCC-BD-FID: Multi-Objective Clustering Combination-Based Backdoor Defense for Federated Intrusion Detection of Industrial Control Systems

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Abstract—Deep learning and federated learning (FL) play a crucial role in ensuring the security of industrial control systems (ICSs), but they also face severe security threats, especially the threat of backdoor attacks. Most FL backdoor defense methods primarily focus on a single clustering strategy, resulting in low true positive rates (TPR) and true negative rates (TNR) in the attack classification task. Due to the excessive combination scheme of currently available clustering strategies, it is difficult to manually select an appropriate combination scheme of clustering strategies to defense backdoor attacks in federated ICSs. This work is the first time to automatically design a multi-objective clustering combination-based backdoor defense for federated intrusion detection in ICSs, called MoCC-BD-FID. The automated design issue of clustering strategies combination for backdoor defense is formulated as a mixed-variable multi-objective optimization problem, which considers both combinatorial variables, i.e., the combination length and the specific combination of clustering strategies, and continuous variables, i.e., the confidence levels of each combined clustering as the decision variables, and considers maximization of both TPR and TNR as the two objectives. To describe and evolve the

different combinations of 12 clustering strategies with confidence levels, we develop an efficient mixed and variable-length encoding mechanism, and the specifically tailored crossover operation and mutation operation under the framework of nondominated sorting genetic algorithm II. The experiments are conducted on the three widely-used ICS datasets including Secure Water Treatment, Water Distribution, and Power System Attack datasets under two different backdoor attacks. The experimental results demonstrate that MoCC-BD-FID outperforms the single clustering strategy-based backdoor defense methods and five existing backdoor defense methods, i.e., Krum, Weak-DP, FoolsGold, DeepSight, and CrowdGuard, in terms of the classification accuracy of the poisoned model on regular samples and backdoor samples, TPR, and TNR.

Index Terms—Industrial control systems, federated intrusion detection, backdoor defense, automated clustering combination, multi-objective optimization.

I. INTRODUCTION

INDUSTRIAL control systems (ICSs) are vital for managing and automating essential infrastructure, including smart grids, water treatment plants, and manufacturing operations [1], [2]. However, ICSs are highly susceptible to cyber-attacks due to their increased connectivity with information technology networks and the adoption of Internet of Things devices [3]. Recently, a variety of increasingly severe cyber-attacks on ICSs have been reported [4], [5]. These attacks have not only led to significant financial losses for affected companies and entities, but also posed serious risks to the safety of local people. Consequently, the security issue of ICS has garnered considerable attention from the research community.

To guarantee the safe operations of a specific ICS, an intrusion detection system (IDS) is often implemented for real-time monitoring [6]. If any intrusions are detected within the system, the IDS triggers an alert and initiates defensive measures against both known and unknown cyber-attacks. Due to its strong fitting and learning abilities, the deep learning model has emerged as the top choice for most IDSs [7]. Furthermore, federated learning (FL) has gained widespread adoption across various domains as an effective solution for addressing data privacy concerns [8], [9]. Therefore, some recent works have prioritized integrating FL with deep learning for the intrusion detection [10], [11], [12]. Utilizing FL in IDSs not only safeguards data privacy and security but also delivers high classification accuracy for cyber-attacks [13], [14].

Received 5 November 2024; revised 24 May 2025 and 26 June 2025; accepted 1 July 2025. Date of publication 7 July 2025; date of current version 10 July 2025. This work was supported in part by Zhejiang Provincial Natural Science Foundation of China under Grant LZ25F030007; in part by the National Natural Science Foundation of China under Grant 61972288, Grant 62403122, and Grant 92067108; in part by the Key-Area Research and Development Program of Guangdong Province under Grant 2020B0101090004; in part by Shanghai Sailing Program under Grant 24YF2701300; in part by the Natural Science Foundation of Guangdong Province under Grant 2021A151501131; in part by the Ministry of Industry and Information Technology (MIIT) Project Industrial Internet Identification Resolution System Security Monitoring and Protection under Grant TC220H078; and in part by Guangdong Key Laboratory of Data Security and Privacy Preserving, National Joint Engineering Research Center of Network Security Detection and Protection Technology. The associate editor coordinating the review of this article and approving it for publication was Dr. Daisuke Mashima. (Corresponding authors: Guo-Qiang Zeng; Kang-Di Lu.)

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Although FL provides an effective method for protecting data privacy, it simultaneously presents some challenges in ensuring security [15]. Because FL allows federated training without compromising data privacy, the heterogeneous distribution of data among federated clients makes it difficult for the server to verify the integrity of local model updates. This vulnerability makes the IDSs deployed in ICSs based on FL susceptible to backdoor attacks [16], [17], [18], [19]. For instance, during the federated training phase, malicious clients can tamper with the training data or model updates in specific ways, and finally upload the poisoned updates to the server. When the server aggregates these malicious updates, it can lead to a global model with a backdoor, thereby forming a backdoor attack. Therefore, it is urgent to design effective backdoor defense schemes for federated intrusion detection of ICSs.

In response to the new threats posed by backdoor attacks, several clustering strategy-based backdoor defense methods have been proposed. These techniques aim to enhance the security of deep learning models, reduce the impact of backdoor attacks, and improve the robustness of models. It is worth noting that most methods use a single clustering strategy to cluster the processed client update parameters [20], [21] with low true positive rates (TPR) and true negative rates (TNR) in the attack classification task. Only a few methods adopt a combination of two clustering strategies, but these methods were designed by trial and error [22], [23]. Due to the low interpretability of deep learning models in a federated framework, the updated parameters of the model occurring at the client side are generally complex and difficult to interpret. Even after operations such as pruning, dimensionality reduction, and similarity calculation are performed on the model update parameters by using lower dimensions to represent these update parameters, there are still significant challenges in accurately clustering updated parameters. Using various advanced clustering strategies can effectively address this issue. Therefore, clustering results of these update parameters largely depend on whether the selected clustering strategy can accurately perform clustering, which is the most critical step in backdoor defense. Determining the applicability of the selected clustering strategy is often by experimentally comparing its test performance with other clustering strategies [24]. However, when it is necessary to select two or more clustering strategies for combination, experimentally comparing is clearly no longer applicable due to the excessive variety of combination scheme of clustering strategies.

In order to employ the combination of various clustering strategies with confidence levels to defend against backdoors, we establish it as a mixed-variable multi-objective optimization problem. Due to the population-based characteristics, multi-objective evolutionary algorithms, e.g., nondominated sorting genetic algorithm II (NSGA-II) [25], are naturally well-suited for solving multi-objective problems. However, evolutionary algorithms do not possess efficient techniques for generating offspring to address the mixed-variable multi-objective optimization problem. Some evolutionary operation frequently used in continuous problems, e.g., simulated binary crossover [26] and differential evolution mutation [27], can

be introduced to solve the integer and categorical variables after minor adjustments. In [28], a rounding operation was considered based on continuous variable operators. For combinatorial optimization problems, offspring generation methods must be customized to fit the unique characteristics of each problem. Yet, these approaches often lack generalizability and cannot be easily applied to solving various problems. Therefore, for mixed-variable multi-objective clustering combination optimization problem, we need to carefully design coding mechanism, crossover operation, and mutation operation to obtain more effective backdoor defense strategy.

Based on the abovementioned considerations, this paper proposes a multi-objective clustering combination-based backdoor defense for federated intrusion detection of ICSs, called MoCC-BD-FID. The main idea behind this method firstly formulates the combination scheme of clustering strategies with confidence levels for effective backdoor defense of IDSs in ICSs as a mixed-variable multi-objective optimization problem with both combinatorial and continuous decision variables by optimizing the trade-off between TPR and TNR. Then, we solve the established problem efficiently by designing a specifically tailored evolutionary algorithm under NSGA-II framework [25].

The main contributions of this work are summarized as follows:

- (1) Most federated IDSs for ICSs focus on detecting cyber-attacks and do not consider the possibility of backdoor attacks in the system. Additionally, most clustering strategy-based backdoor defense methods typically use a single or two types of clustering strategies and contain the following shortcomings: low performance in TPR and TNR and lack an effective automatic selection mechanism to choose different combination schemes when involving multiple clustering strategies. This work originally develops a backdoor defense method through the automated design combination scheme of 12 different clustering strategies for federated intrusion detection of ICSs. To the best of our knowledge, this paper is the first to automatically design the backdoor defense method for federated intrusion detection of ICSs from the perspective of multi-objective clustering combination.

- (2) In the design process, the defense strategy against backdoor attacks for federated intrusion detection of ICSs utilizing a combination of different clustering strategies is formulated as a mixed-variable multi-objective optimization problem, where maximization of both TPR and TNR is defined as the two objective functions. Furthermore, we develop an efficient mixed-variable and variable-length encoding scheme, and the specified crossover and mutation operations under the framework of NSGA-II to describe and evolve various combinations of 12 classic clustering strategies with confidence levels, i.e., the proportion of each clustering strategy in the classification result.

- (3) The experimental results on three ICS intrusion datasets including Secure Water Treatment (SWaT) [29], Water Distribution (WADI) [30], and Power System Attack (PSA) [31] datasets indicate that the MoCC-BD-FID obtains satisfactory defense performance against two different backdoor attacks, i.e., SIG [32] and BadNet [33]. Moreover, the

proposed MoCC-BD-FID outperforms other 12 types of single clustering strategy-based backdoor methods and five existing backdoor defense methods including three classical methods, i.e., Krum [34], Weak-DP [35], FoolsGold [36] and two manually designed methods, i.e., DeepSight [37] and CrowdGuard [22], in terms of the classification accuracy of the poisoned model on regular samples and backdoor samples, TPR, and TNR.

II. RELATED WORKS

In order to effectively address the challenges posed by backdoor attacks in FL, various defense methods were presented [21], [22], [23], [24], [38], [39]. Zhang et al. [21] presented SecFedNIDS, which first provided an efficient low-dimensional representation of the client local model update parameters and then used the Stochastic Outlier Selection (SOS) strategy to cluster these parameters, thereby identifying and filtering out malicious update parameters. Rieger et al. [22] proposed federated backdoor detection method, called CrowdGuard. It utilized feedback from clients regarding specific models and removed compromised models using a step-by-step pruning approach. Additionally, CrowdGuard implemented a stacked clustering method on the server side to strengthen its resistance against malicious client input. In [23], RDFL was proposed. In RDFL, appropriate update parameters were selected to compute the cosine distance. HDBSCAN and Hierarchical Clustering strategies were combined to perform adaptive clustering to detect and remove suspicious malicious client updates, followed by adaptive pruning and noise addition operations. Nguyen et al. [24] proposed the FLAME defense framework. It first measured the angular differences between all model update parameters by calculating the cosine distance of the update parameters from each client model in pairs. Then, it clustered these update parameters based on the dynamic HDBSCAN clustering strategy to identify and filter out malicious updates. Finally, in the server-side parameter aggregation phase, it effectively eliminated backdoors by dynamically pruning the update parameters and adding sufficient noise. In [37], a model filtering method, called DeepSight, was presented. DeepSight employed three novel techniques to analyze the training data distribution and detect subtle differences in neural networks, enabling it to identify suspicious model updates. By clustering these updates and applying weight clipping defenses, it effectively mitigated the impact of potential backdoor contributions from undetected poisoned models. Chen et al. [39] presented RFBDS method and the privacy-preserving version, i.e., PrivRFBDS to ensure the elimination of malicious backdoors. RFBDS first used Amplified Magnitude Sparsification to extract the major client local model update parameters and then calculated the median Euclidean distance between the model updates. After that, it performed clustering based on the OPTICS or DBSCAN strategy and finally executed an adaptive model pruning operation to remove existing backdoors.

Most methods use a single clustering strategy to cluster the processed client update parameters, while only a few methods combined two clustering strategies. Due to the low interpretability of deep learning models, the model update

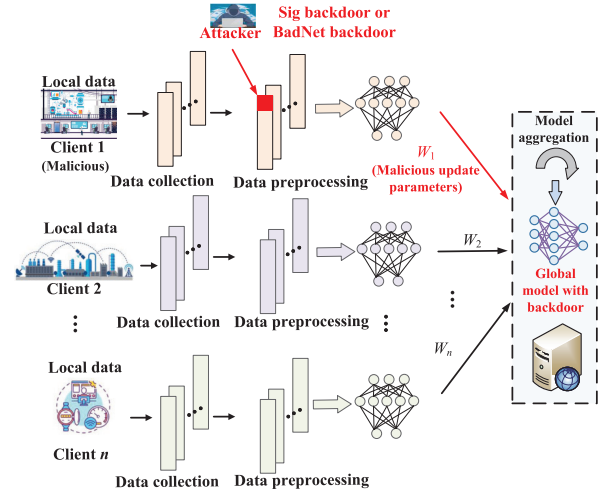


Fig. 1. The process of backdoor attack in FL.

parameters from clients are complex and difficult to interpret. Even after pruning, dimensionality reduction, and similarity calculations of the model update parameters, accurately clustering these parameters remains a significant challenge. Although there are backdoor defense methods based on two clustering strategies, these methods are obtained through manually designed based on the extensive experience of experts, making it difficult to apply when multiple clustering strategies are involved.

III. THE PROPOSED MOCC-BD-FID METHOD

A. Assumptions

1) *Attacker's Goals*: The goal of the attacker is to compromise the global intrusion detection model. The attacker manipulates the controlled clients to perform local training and sends malicious local update parameters to the server. During the server's parameter aggregation process, a specific backdoor is implanted into the global model. This backdoor causes the global model to misjudge specific traffic types, such as classifying malicious traffic as normal, thereby facilitating attacks on ICSs. The process of the backdoor attack in FL is illustrated in Fig. 1.

2) *Attacker's Strategy*: The attacker considers two types of backdoor attacks against FL, called the SIG backdoor attack [32] and the BadNet backdoor attack [33]. Both types of attacks are simple and efficient to implement in real-world settings. The SIG attack uses an overlaid sinusoidal signal as a trigger, while BadNet uses a small square with a maximum pixel value of 255 as a trigger. These methods are applied to traffic to construct backdoor samples while adhering to certain constraints in ICSs.

B. The Process of Backdoor Defense

In the scenario of defending against backdoor attacks in federated intrusion detection of ICS, it is crucial to process the updated parameters uploaded by clients. Typically, each update parameter contains a large number of features, including a lot of redundant information. To reduce this redundant information and lower the complexity of subsequent computations, it

is necessary to first perform dimensionality reduction on the update parameters. Here, principal component analysis (PCA) is used to perform dimensionality reduction on the update parameters shown in (1) as follows:

$$W = \begin{bmatrix} W_{11} & \dots & W_{1n} \\ W_{21} & \dots & W_{2n} \\ \dots & \dots & \dots \\ W_{m1} & \dots & W_{mn} \end{bmatrix} \rightarrow V = \begin{bmatrix} V_{11} & \dots & V_{1k} \\ V_{21} & \dots & V_{2k} \\ \dots & \dots & \dots \\ V_{m1} & \dots & V_{mk} \end{bmatrix} \quad (1)$$

where W represents the set of updated parameters from all clients participating in training under the federated framework. m is the number of clients, with each row representing the update parameters of a single client. n denotes the number of features included in each update parameter. V is the value after PCA and k is the number of features contained in each update parameter after dimensionality reduction after PCA.

For the updated parameters V after dimensionality reduction, pairwise cosine distances are calculated to obtain the cosine distance matrix H shown in (2)-(3) as follows:

$$H = \begin{bmatrix} \cos_{11} & \cos_{12} & \dots & \cos_{1m} \\ \cos_{21} & \cos_{22} & \dots & \cos_{2m} \\ \dots & \dots & \dots & \dots \\ \cos_{m1} & \cos_{m2} & \dots & \cos_{mm} \end{bmatrix} \quad (2)$$

$$\cos_{uv} = \cos(V_u, V_v) = 1 - \frac{V_u \times V_v}{\|V_u\| \|V_v\|}, 1 \leq u, v \leq m \quad (3)$$

It is worth noting that one advantage of calculating cosine distance is that even if a malicious client scales up the local model's updated parameters to enhance its impact, the cosine distance remains unaffected because the angle between the update weight vectors is not changed.

Using the optimized combination scheme of clustering strategies obtained by MoCC-BD-FID method to cluster the cosine distance matrix H . Suppose the optimized combination scheme is the $Cluster = [l, (x_1, cl_1), \dots, (x_j, cl_j), \dots, (x_l, cl_l)]$, then the classification results of all clustering strategies are combined to obtain the final result of each updated parameter. The specific calculation process is given in (4) as follows:

$$result = \sum_{j=1}^l x_j \times cluster_j = [r_1, r_2, \dots, r_h, \dots, r_m] \quad (4)$$

where l is the encoding length. cl_j represents the selected clustering strategy. $cluster_j$ is the classification result of the clustering strategy on the cosine distance matrix. x_j denotes the confidence level of $cluster_j$. For the clustering result of a single clustering strategy, the category with the highest proportion is labeled as 0, meaning a normal update, while all other categories are labeled as 1, indicating an abnormal update. If there are multiple categories with the highest proportion, one category is randomly selected to be labeled as 0, and the remaining categories are labeled as 1.

Set the classification threshold th , and classify the result of the combined clustering $result$ to obtain the final classification $predict$ shown in (5) as follows:

$$predict = [pr_1, pr_2, \dots, pr_i, \dots, pr_m] \\ pr_i = \begin{cases} 0, r_h < th \\ 1, r_h \geq th, \end{cases} \quad 1 < h < m \quad (5)$$

Finally, select the updated parameters from the classification results of 0 in the prediction. Carry out server-side parameter aggregation and then redistribute the aggregated parameters to each client.

C. Problem Formulation

In this paper, the searching for an optimized combination scheme of clustering strategies is formulated as a mixed-variable multi-objective optimization problem. The detailed decision variables, i.e., $X = [l, x_1, cl_1, \dots, x_j, cl_j, \dots, x_l, cl_l]$ with two objective functions $F(X)$ are given as follows:

$$\begin{aligned} \max \quad & F(X) = (f_1(X), f_2(X)) \\ \begin{cases} f_1 = \text{TPR} = TP/(TP + FN) \\ f_2 = \text{TNR} = TN/(TN + FP) \end{cases} \\ \text{s.t.} \quad & \sum_{j=1}^l x_j = 1, [cl_1, \dots, cl_j, \dots, cl_l] \in \text{ClusterSet}, 1 \leq l \\ & \leq 12 \end{aligned} \quad (6)$$

where

$$\begin{aligned} [TP, TN, FP, FN] &= f(W_{te}, label_{te}, X^*) \\ X^* &= \arg \min_X MSE(W_{tr}, label_{tr}, X) \\ MSE &= \frac{1}{N} \sum_{i=1}^N (l_i - r_i)^2 \\ label_{tr} &= [l_1, l_2, \dots, l_i, \dots, l_N] \\ result &= [r_1, r_2, \dots, r_i, \dots, r_N] \\ &= \sum_{j=1}^l (x_j \times cl_j(W_{tr})) \end{aligned} \quad (7)$$

where f_1 and f_2 represent two objective functions to obtain TPR and TNR. It should be noted that the TPR and TNR mentioned above are metrics used to evaluate the effectiveness of clustering methods. Here, TNR represents the proportion of normal gradients correctly identified as normal in the clustering results. Generally, we aim for higher TNR values because only gradients classified as normal are aggregated to update the global model. A higher TNR indicates fewer malicious gradients being incorporated into the global model. TPR measures the proportion of malicious gradients correctly identified as malicious in the clustering results. These definitions differ from the TPR and TNR metrics used in conventional model performance evaluation. TP represents the number of malicious samples correctly classified as intrusions. TN represents the number of normal samples correctly classified as normal. FP represents the number of normal samples incorrectly classified as intrusions. FN represents the number of malicious samples incorrectly classified as normal. W_{tr} is the training set and W_{te} is the test set, respectively. $label_{tr}$ and $label_{te}$ are the labels correspond to W_{tr} and W_{te} , respectively, indicating the nature of the updates: 0 for normal updates and 1 for malicious updates. l is the length of the selected clustering strategies. $[cl_1, \dots, cl_j, \dots, cl_l]$ is a combination scheme of clustering strategies, containing a free combination scheme of 12 different clustering strategies. **ClusterSet** is the set of 12 different clustering strategies. $[x_1, \dots, x_j, \dots, x_l]$ denotes the confidence level for each clustering strategy in

TABLE I
THE INDEX AND CORRESPONDING CLUSTERING STRATEGY

Index	Clustering Strategy	Index	Clustering Strategy
1	HDBSCAN	7	Gaussian Mixture
2	DBSCAN	8	MiniBatch KMeans
3	K-Means	9	Birch
4	MeanShift	10	Agglomerative Clustering
5	OPTICS	11	Affinity Propagation
6	Spectral Clustering	12	SOS

the combination. The sum of all confidence levels equals 1. $cl_j(W_{it})$ represents the clustering result of a single clustering strategy on the update parameters within each sample group. $result$ is the result of clustering based on the combination scheme. X^* is the optimal X according to the minimization of mean squared error (MSE). N is the number of samples. f is the relationship of testing result to obtain TP , TN , FP , and FN based on W_{te} , $label_{te}$, and X^* .

Table I gives the index and the corresponding clustering strategy. The details of 12 clustering strategies are described as follows:

1) *HDBSCAN*: HDBSCAN is a density-based clustering strategy aiming to identify clusters with varying densities and can handle noise points. HDBSCAN combines the advantages of DBSCAN and hierarchical clustering, effectively dealing with noise and outliers in the data.

2) *DBSCAN*: DBSCAN is a density-based clustering strategy used to identify clusters with high density and effectively handles noise points. DBSCAN does not require the number of clusters to be specified in advance and can adaptively identify clusters of arbitrary shapes. It is insensitive to the choice of parameters.

3) *K-Means*: K-Means is a common clustering strategy used to divide data points into K different clusters, aiming to minimize the squared distance between each data point and the center of its respective cluster. K-Means accomplishes the clustering process by iteratively optimizing the positions of the cluster centers.

4) *MeanShift*: MeanShift is a density-based nonparametric clustering strategy used to discover density estimate patterns in data, especially suited for identifying clusters of arbitrary shapes. MeanShift continuously adjusts the positions of data points, moving them towards the local maxima of the data density, thereby locating the centers of the clusters.

5) *OPTICS*: OPTICS is a density-based clustering strategy designed to identify cluster structures and noise points in data. Similar to the DBSCAN, OPTICS can also handle clusters of arbitrary shapes and does not require the pre-specification of the number of clusters.

6) *Spectral Clustering*: Spectral Clustering is a spectral clustering strategy based on graph theory and spectral theory groups data by transforming it into a low-dimensional feature space and performing clustering in space. Spectral clustering strategies are typically used to handle clustering problems involving non-spherical and linearly inseparable data.

7) *Gaussian Mixture*: Gaussian Mixture is a probabilistic model used to describe data distributions formed by the combination of multiple Gaussian distributions, typically used

for clustering analysis and density estimation problems. The basic assumption is that the data is a mixture of multiple Gaussian distributions.

8) *MiniBatch KMeans*: MiniBatch KMeans is a variant of the K-Means strategy aiming to accelerate the computation speed of the K-Means strategy, making it particularly suitable for handling large-scale datasets. Unlike the traditional K-Means strategy, which processes all data points at once, the MiniBatchKMeans strategy updates cluster centers using small batches, thereby reducing computational complexity.

9) *Birch*: Birch is a hierarchical clustering strategy for large-scale datasets. It aims to cluster efficiently by gradually reducing the dimensions and density of the dataset, making it suitable for handling large amounts of data while being constrained by memory.

10) *Agglomerative Clustering*: Agglomerative Clustering is a hierarchical clustering strategy. Agglomerative Clustering starts from each data point, treating each data point as an individual cluster, and then gradually merges adjacent clusters until the stopping condition is met.

11) *Affinity Propagation*: Unlike traditional clustering strategies based on distance or density, the Affinity Propagation strategy determines the propagation weights between data points by calculating the similarity between them, thereby achieving clustering.

12) *SOS*: SOS is an anomaly detection algorithm based on random sampling, which is used to select outliers or anomalous values in a dataset. SOS identifies potential outliers by randomly selecting samples and calculating similarity to other samples.

D. The Framework of MoCC-BD-FID Method

In this work, we propose a clustering combination optimization approach that combines multiple clustering algorithms to enhance the robustness of detection. This optimization problem is inherently a mixed-variable multi-objective optimization problem. NSGA-II excels in such scenarios due to their natural selection mechanisms, which enable global search capabilities without relying on continuous differentiability.

In addition, NSGA-II inherently supports mixed-variable encoding and parallel computation, accelerating convergence for mixed-variable multi-objective optimization problem. In contrast, reinforcement learning requires extensive environmental interaction, particularly suffering from action space explosion in mixed-variable settings, leading to high training costs. Sparse and delayed reward signals further complicate policy exploration in multi-objective optimization. Bayesian optimization depends on prior distribution modeling with insufficient priors or sparse data, convergence slows and sub-optimal solutions. Deep learning models, primarily designed for classification and regression, struggle to directly handle mixed-variable multi-objective optimization. Therefore, we use NSGA-II as the main optimization tool to solve the mixed-variable multi-objective optimization problem.

The diversity of clustering algorithms results in an enormous number of possible combinations. For instance, simple permutations and combinations of 12 fundamental methods already yield over 4,000 candidate solutions. Furthermore,

Algorithm 1 Framework of the Proposed MoCC-BD-FID Method

Input: Population size N , the maximum number of generation G , crossover probability θ_c , mutation probability θ_m , and the set of 12 clustering algorithms $ClusterSet$.

Output: The Pareto front of backdoor defense methods based on combination scheme of clustering strategies.

- 1 $S_0 \leftarrow$ Use the **developed encoding scheme** with $ClusterSet$ and randomly initialize a population containing N individuals;
- 2 Evaluate the fitness of all individuals in S_0 ;
- 3 $g \leftarrow 0$;
- 4 $S_n \leftarrow S_0$;
- 5 **while** $g < G$ **do**
- 6 $Q_g \leftarrow$ Generate offspring population for the population S_n by the **developed crossover and mutation operations** with θ_c and θ_m ;
- 7 Evaluate the fitness of all individuals in Q_g ;
- 8 $S_n \leftarrow$ Obtain the next population by performing environmental selection on $S_n \cup Q_g$;
- 9 $g \leftarrow g + 1$;
- 10 **end**
- 11 **Return** The Pareto front of backdoor defense methods based on the combination scheme of clustering strategies.

the introduction of clustering method confidence parameters significantly increases complexity, causing the number of candidate solutions to grow exponentially. Under these circumstances, exhaustive enumeration becomes computationally infeasible. Therefore, to efficiently identify optimal solutions, we employ NSGA-II that automatically optimize strategies to balance search breadth with computational cost, evaluating and selecting the best clustering combination strategy through objective function assessment.

The overall framework of the proposed MoCC-BD-FID method is given in Fig. 2. The pseudocode of MoCC-BD-FID method is shown in Algorithm 1. The proposed MoCC-BD-FID method consists of four main parts: population initialization, fitness calculation, offspring generation and environmental selection. First, determine the related input parameters including population size N , the maximum number of generation G , crossover probability θ_c , mutation probability θ_m , and the set of 12 clustering strategies $ClusterSet$. Use the developed encoding scheme to randomly initialize the parent population S_0 containing N individuals and evaluate the fitness of all individuals in S_0 . Next, perform the developed crossover operation and mutation operation on the parent population S_0 to generate the offspring population Q_g , and evaluate the fitness of all individuals in Q_g . Merge the parent population S_n and offspring population Q_g for environmental selection to obtain the next generation population and update S_n . Repeat the above operations until the G are reached. Obtain the Pareto front in the final population as the corresponding combination scheme of clustering strategies and the confidence

Algorithm 2 Fitness Calculation

Input: Population to be evaluated S_n , training dataset W_{tr} , test dataset W_{te}

Output: The fitness of S_n .

- 1 **for** each individual $Indi_i$ in S_n **do**
- 2 $Cluster_i \leftarrow$ According to the encoded information in individual $Indi_i$, obtain corresponding combination scheme and confidence levels;
- 3 Optimize the confidence of combination scheme in $Cluster_i$ using W_{tr} ;
- 4 Test the optimized combination scheme $Cluster_i$ using W_{tr} ;
- 5 $Fitness \leftarrow$ Calculate the average TPR and TNR;
- 6 Assign $Fitness$ to $Indi_i$;
- 7 **end**
- 8 **Return** The fitness of S_n .

level corresponding to each clustering strategy, and select one solution in Pareto front for backdoor attack defense in the FL-based intrusion detection system of ICS.

We employ NSGA-II to obtain clustering combination strategy for the cosine similarity matrix clustering. In practical deployment, the MoCC-BD-FID is conducted offline. During online detection, the optimized clustering scheme is directly applied without introducing additional latency, thus satisfying real-time response requirements. Furthermore, since FL model aggregation is inherently periodic, the MoCC-BD-FID only needs to be executed once during the initial federated training phase. Subsequent model update cycles do not require multiple iterations.

E. Population Initialization

Initialize N individuals as the initial population $S_0 = \{Indi_i, i = 1, 2, \dots, N\}$, where $Indi_i$ represents the encoding of the i -th individual. Each individual represents a combination scheme of clustering strategies for backdoor defense. The encoding format of each individual in the population is $Indi_i = [l_i, (x_{i1}, c_{i1}), \dots, (x_{ij}, c_{ij}), \dots, (x_{ili}, c_{ili})]$, where l_i represents the length of the current individual encoding. The maximum length of an individual encoding is 12 and the minimum length is 1. c_{ij} represents the j -th encoding in the i -th individual, where $1 \leq j \leq l_i$. c_{ij} is an integer between 1 and 12, corresponding to the serial number of a selected clustering strategy. x_{ij} represents the confidence level of the clustering strategy c_{ij} , with values ranging from 0 to 1. Fig. 3 shows an example of mixed-variable encoding scheme of clustering strategy.

F. Fitness Calculation

The pseudocode for fitness calculation of MoCC-BD-FID is shown in Algorithm 2. For each individual $Indi_i$ in the population S_n , decode the $Indi_i$ and construct the combination scheme of clustering strategies to form a backdoor defense. The combination scheme is evaluated using each sample set from the training set. For each sample set W , we can calculate H and $result$ according to (1)–(3)

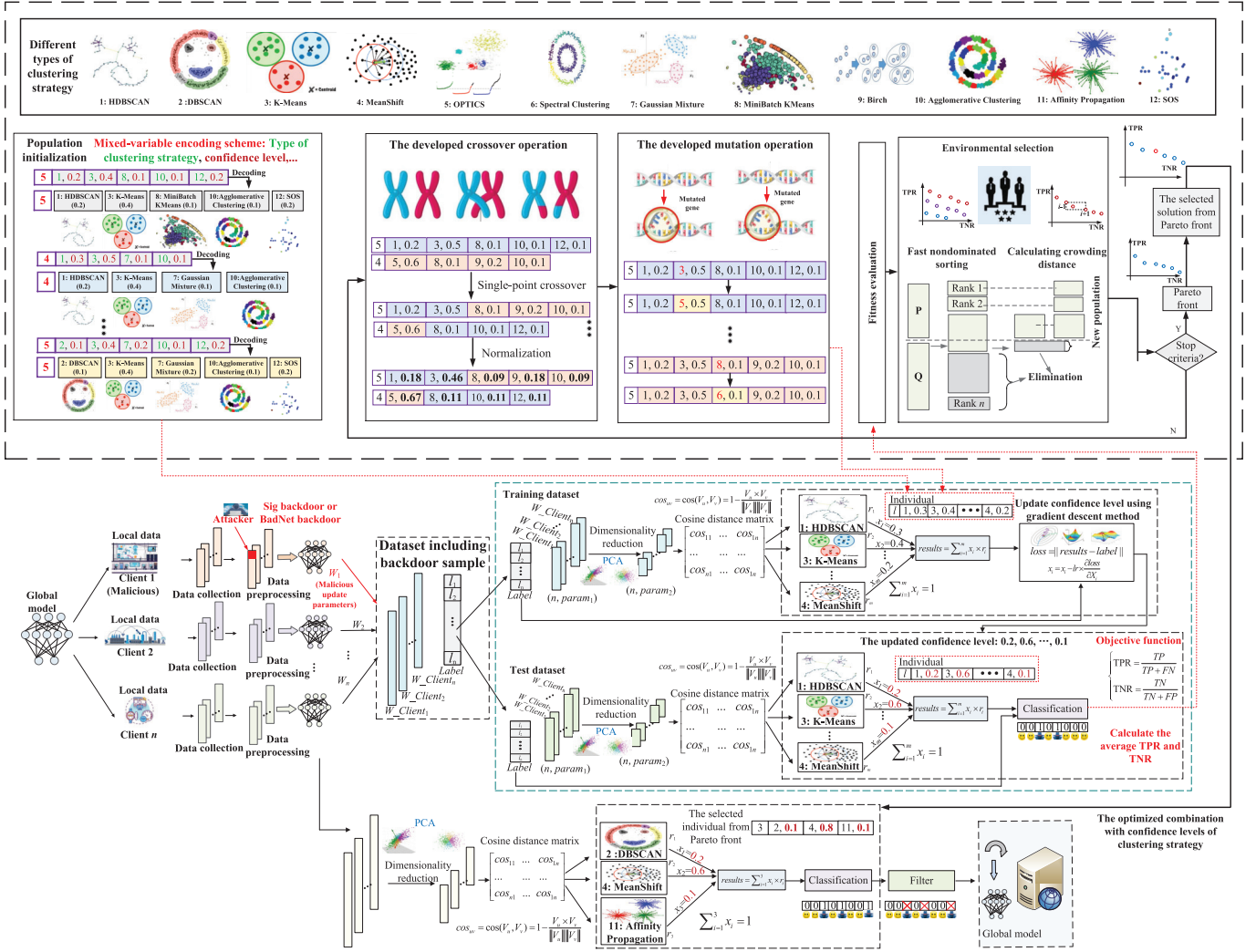


Fig. 2. The overall framework of the proposed MoCC-BD-FID method.

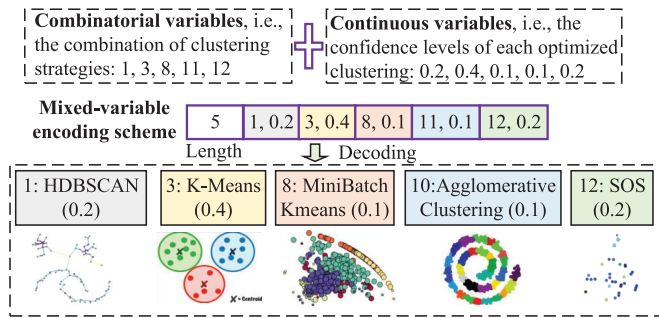


Fig. 3. An example of mixed-variable encoding scheme of clustering strategy.

Calculate the loss between the *result* and the true label $label_{tr} = [l_1, l_2, \dots, l_N]$, as shown in (8):

$$loss = \frac{1}{N} \sum_{i=1}^N (l_i - r_i)^2 \quad (8)$$

Optimize the confidence coefficient x_i through the gradient descent method, as shown in (9)

$$x_i = x_i - l_r \times \frac{\partial loss}{\partial x_i}, 1 \leq i \leq l \quad (9)$$

where l_r is the learning rate. For the updated x_i , perform normalization according to (10)–(11) as follows:

$$sumx = \sum_{i=1}^l x_i, i = 1, 2, \dots, l \quad (10)$$

$$normx_i = \frac{x_i}{sumx}, i = 1, 2, \dots, l \quad (11)$$

The final confidence level x_i is optimized based on all samples in the training set. Then, the combination scheme is tested using the test set, which is composed similarly to the training set. For each sample w_{te} in the test set, the clustering results are classified according to (1)–(4). For each sample in the test set, based on the true label $label_{te}$ of each sample, TPR and TNR are calculated as the fitness of the current combination scheme.

G. Offspring Generation

The population S_n undergoes crossover and mutation operations to produce the offspring population Q_g . Two individuals are randomly selected from S_n and denoted as $Indi_1$ and $Indi_2$, respectively. A random real number r_c within the range (0,1) is generated. If $r_c < \theta_c$, a position is randomly selected from

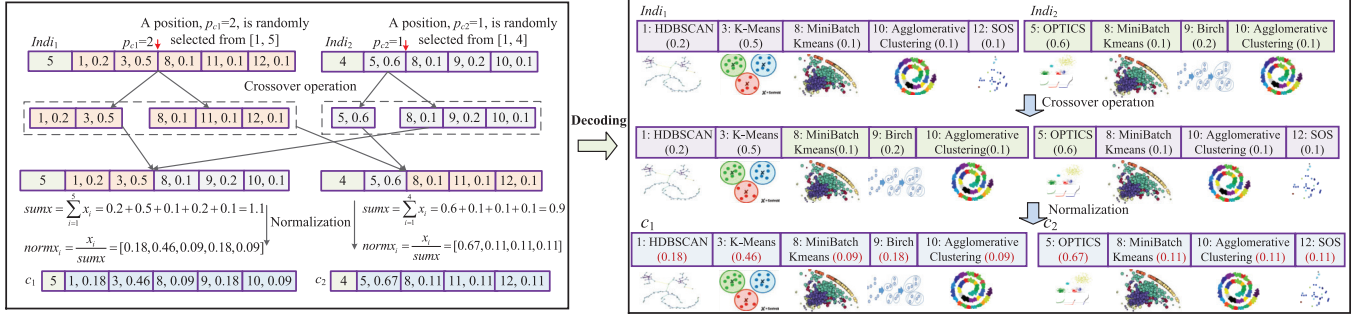


Fig. 4. An example of the developed crossover operation.

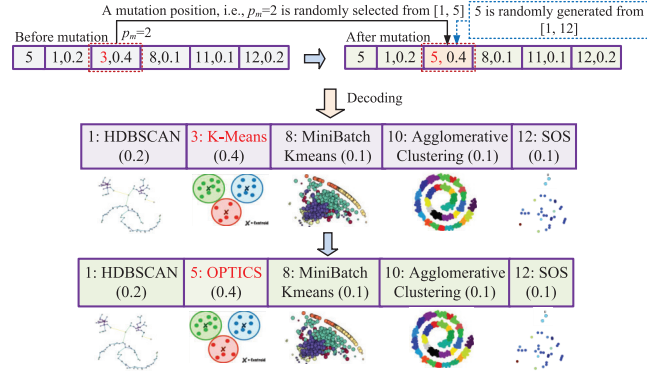


Fig. 5. An example of the developed mutation operation.

$Indi_1$ and $Indi_2$ as the crossover point, and a single-point crossover is performed on $Indi_1$ and $Indi_2$ to produce two offspring individuals, i.e., c_1 and c_2 . If $r_c \geq \theta_c$, then $c_1 = Indi_1$ and $c_2 = Indi_2$.

Mutation operation is then performed on the offspring after crossover. For each individual c_i , a random real number r_m within the range (0,1) is generated. If $r_m < \theta_m$, a position is randomly selected from c_i as the mutation point. An integer p_i within the range from 1 to 12 that is different from the initial value at the mutation point is randomly generated, replacing the value at the mutation point in c_i , thereby producing a mutated individual.

The above-mentioned crossover operation and mutation operation are repeated until the number of individuals in the offspring population Q_g reaches N . Fig. 4 and Fig. 5 provide an example of the crossover operation and mutation operation, respectively.

H. Environmental Selection

During the environmental selection phase, the framework of NSGA-II [25] is used for environmental selection. First, the parent population and the offspring population are merged, and a fast non-dominated sorting is performed on the combined population. Then, the individuals in the population are ranked based on dominance and non-dominance relationship and the crowding distance. Finally, an elitist strategy is adopted to select N individuals as the next generation population.

IV. EXPERIMENTAL RESULTS

The experiments are carried out on a server equipped with an Intel® Xeon® E5-2683v3 @ 2.00 GHz processor, 256 GB

of memory, and 4 A100-PCIE graphics cards. Binary supervised learning tasks are conducted by using SWaT, WADI, and PSA datasets that contain normal and attack samples. For dataset processing, Min-Max normalization is applied to the SWaT, WADI, and PSA datasets.

During the experiments, we configure 100 federated client nodes, with each client's training data maintaining independent and identically distributed characteristics to the greatest extent possible. Four A100 GPUs are allocated to these simulated clients to provide computational resources for data processing and federated training. Among these 100 federated clients, we randomly selected 10 clients as malicious participants, whose training datasets contained a small proportion of backdoor attack samples. Each client performed local training to generate either normal or backdoor-embedded gradients for server-side parameter aggregation. The GPU acceleration significantly reduces the overall federated training duration, thereby shortening running time. Additionally, the NSGA-II primarily relied on CPU computation, where GPU acceleration showed limited performance improvement.

A. Datasets

(1) SWaT [29]: The SWaT (Secure Water Treatment) dataset originates from physical testbed systems for water treatment managed by Singapore's Public Utility Board. Within the field of cyber-security, SWaT is frequently utilized to examine the impact of cyber-attacks on ICSs, assess the efficacy of intrusion detection techniques, and measure the resilience of defense mechanisms under cyber-attack scenarios. This dataset monitors 51 different variables and includes data reflecting various operational conditions, such as normal operations and attack situations.

(2) WADI [30]: WADI (Water Distribution) is an expanded version of the SWaT system. It represents a distribution network made up of numerous water distribution pipelines, making it a more intricate and practical water treatment system. The dataset tracks 123 variables and encompasses data from both normal operations and attack conditions.

(3) PSA [31]: The PSA (Power System Attack) dataset, developed by Uttam Adhikari and Shengyi Pan from Oak Ridge National Laboratories in USA, integrates measurement data capturing typical operations, equipment malfunctions, and cyberattack behaviors in power grids. It employs four primary data sources: phasor measurement units (PMUs), smart control panel logs, and relay statuses, collectively forming a

TABLE II
THE STATISTICS OF DATASETS

Dataset	# Class	# Dimension	# Samples
SWaT	2	51	449919
WADI	2	123	172801
PSA	2	128	4967

128-dimensional feature set. Each PMU records 29 electrical parameters, which are combined with relay alarms and logs. The dataset also includes 12 control panel log columns and one labeled attack-type column. Here, we use subset 1 from PSA as the case study.

The SWaT and WADI datasets originate from Singapore's ICSs, while SPA dataset comes from Oak Ridge National Laboratories in the USA. Using SPA dataset demonstrates the transferability of the proposed MoCC-BD-FID approach for different ICSs.

Table II gives the statistics of SWaT, WADI, and PSA datasets.

B. Attack Scenarios

First, the effectiveness of backdoor attacks in the datasets of ICSs is validated by generating respective backdoor samples using SIG and BadNet methods, which are then added to the training set for producing malicious update parameters. In the FL setup, the number of malicious clients is set to 10%, with each malicious client's training set containing 5% backdoor samples. As a deep learning-based detection method, 1D-CNN has been widely applied in ICSs and has achieved good results [14]. Thus, we use 1D-CNN model to deploy in the federated IDS.

The WADI dataset serves as an extension of the SWaT dataset, both targeting water treatment-related industrial control scenarios. They share consistent data collection methods, sensor/actuator types, and data format designs, resulting in similar feature semantics. For clarity, we use the SWaT dataset as an example here. In SWaT, each sample contains 51 features, where some features represent sensor readings, i.e., continuous values within specific ranges, and other features indicate controller states, i.e., discrete and limited values, e.g., 0 for "off" and 1 for "on".

When constructing backdoor samples from initial samples, typically normal attack samples, we modify specific feature values to embed backdoors while ensuring compliance with feature characteristics and physical constraints. For SIG backdoor samples, we generate a 51-dimensional discrete low-frequency sine wave signal aligned with SWaT's feature dimensions. Then, superimpose sine wave components only on sensor features while preserving controller values and dynamically adjust certain controller features based on physical system constraints to maintain operational logic. For BadNet backdoor samples, we select 2-4 critical sensors as triggers, assigning fixed values, e.g., median feature values, that satisfy physical constraints. Then, maintain unchanged values for remaining sensors and controllers and ensure trigger consistency across all BadNet samples. To measure the effectiveness of backdoor attacks, we use two metrics: the classification

accuracy of the poisoned model on regular samples (MA) and the classification accuracy of the poisoned model on backdoor samples (BA). The detailed definitions of MA and BA are given in (12) and (13) as follows:

$$MA = \frac{TP + TN}{TP + TN + FP + FN} \quad (12)$$

$$BA = \frac{TN}{TN + FP} \quad (13)$$

To comprehensively evaluate the classification performance, we employ other three metrics including, precision (Pre), recall (Rec), F1-score (F1), and false positive rate (FPR). The detailed expressions of Pre, Rec, and F1 are given as follows:

$$Pre = \frac{TP}{TP + FP} \quad (14)$$

$$Rec = \frac{TP}{TP + FN} \quad (15)$$

$$F1 = \frac{2 \times Pre \times Rec}{Pre + Rec} \quad (16)$$

$$FPR = \frac{FP}{FP + TN} \quad (17)$$

The definitions of TP , TN , FN , and FP have been previously explained in (6).

Table III shows the effectiveness of backdoor attack methods, i.e., SIG and BadNet, without filtering malicious update parameters. As can be seen from Table III, without defense, the SIG and BadNet backdoor attacks can eventually achieve a high success rate after multiple rounds of federated training. Additionally, Fig. 6 presents the confusion matrix of the classification model. From Fig. 6 and Table III, we can see that the model maintains strong classification performance both before and after suffering from backdoor attacks. Fig. 7 presents the results of SIG and BadNet on the SWaT dataset, including the original samples, SIG backdoor attack-based samples, and BadNet backdoor attack-based samples. From Fig. 7, it can be seen that the attack samples generated by the SIG and BadNet method are close to the original samples. Clearly, both SIG and BadNet ensure that the generated attack samples are closely matched to the original samples.

False alarms refer to the misclassification of normal traffic and it can be quantified by the model's FPR metric. Based on the FPR calculations, we can obtain the detailed false alarms per hour. For the SWaT dataset, the FPR ranges from 0.0055 to 0.0063, resulting in 20 to 23 false alarms per hour. For the WADI dataset, the FPR ranges from 0.0010 to 0.0011, yielding 4 false alarms per hour.

The TPR quantifies the system's ability to detect actual attacks, while the TNR reflects system stability, and the FPR represents operational costs associated with false alarms. Although even small FPR values, e.g., 0.01%, can generate significant false alarms in large-scale ICSs, we can develop efficient and standardized alarm rationalization by incorporating specific Industrial Cybersecurity Standards. For instance, we can design a natural language processing model based on the ISA 18.2 industrial alarm management standard to automatically map penetration testing findings to Adversarial Tactics, Techniques, and Common Knowledge relationships,

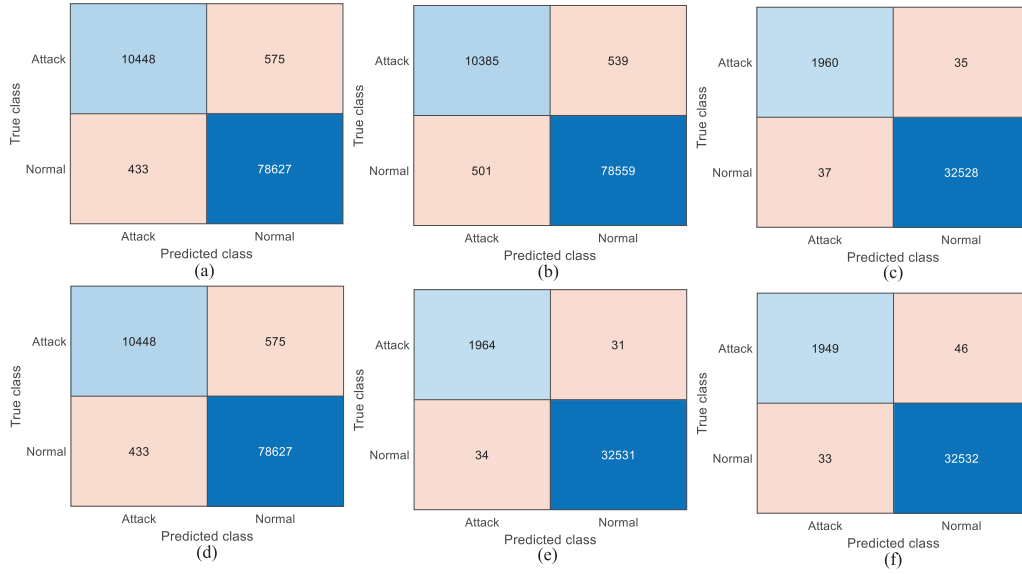


Fig. 6. The confusion matrix of the classification model. (a) SWaT dataset under no backdoor attack; (b) SWaT dataset under SIG attack; (c) SWaT dataset under BadNet attack; (d) WADI dataset under no backdoor attack; (e) WADI dataset under SIG attack; (f) WADI dataset under BadNet attack.

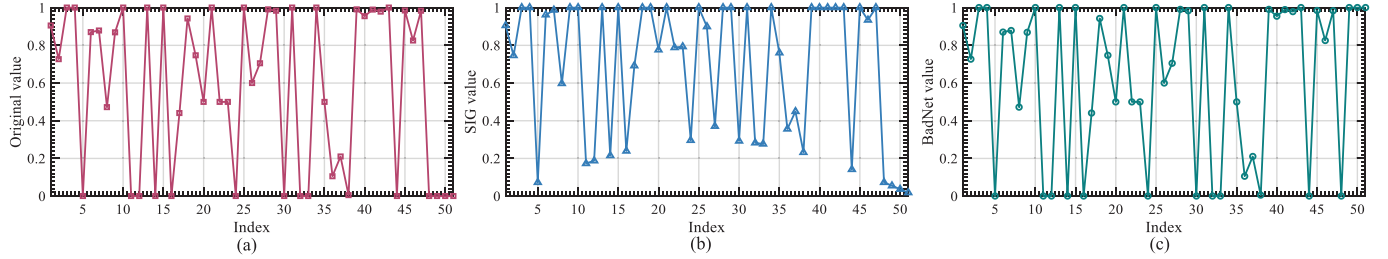


Fig. 7. Results of backdoor attack on SWaT dataset. (a) Original sample; (b) SIG backdoor attack-based sample; (c) BadNet backdoor attack-based sample.

TABLE III

THE CLASSIFICATION PERFORMANCE OF THE MODEL ON THE SWAT AND WADI DATASET UNDER DIFFERENT ATTACK SCENARIOS

Dataset	Method	MA	BA	Pre	Rec	F1	FPR
SWaT	No Backdoor	0.9888	-	0.9602	0.9478	0.9539	0.0055
	SIG	0.9884	0.9559	0.9540	0.9507	0.9523	0.0063
	BadNet	0.9883	0.8621	0.9593	0.9437	0.9514	0.0055
WADI	No Backdoor	0.9979	-	0.9815	0.9825	0.9820	0.0011
	SIG	0.9981	0.9647	0.9830	0.9845	0.9837	0.0010
	BadNet	0.9977	0.8559	0.9834	0.9769	0.9801	0.0010

thereby achieving rationalized alarm management [40]. Furthermore, as suggested in [41], applying MoCC-BD-FID to more realistic training datasets enhances the identification of genuine attacks. Additionally, we proactively address noisy labels that may arise from attack attempts or benign triggers, further improving alarm rationalization. For systems that cannot tolerate any false positives, we can integrate ε -classifiers, adopting zero false positives as the primary decision-making metric [42].

C. Evolutionary Process of MoCC-BD-FID Method

Fig. 8 presents the Pareto fronts of MoCC-BD-FID on the SWaT and WADI datasets under SIG and BadNet backdoor attacks. We provide the Pareto fronts at generations 1, 5, 10, and 20. As shown in Fig. 8, for the SWaT dataset, both the convergence and diversity of the Pareto fronts improve

continuously throughout the evolutionary process. Specifically, the results at the 20th generation are significantly better than those generated randomly at the 1st generation. For the WADI dataset, the quality of the Pareto fronts also improves steadily. Although some solutions are similar in the 5th and 10th generations, the diversity of solutions improves by the 20th generation, which is an important metric for Pareto fronts. The diversity of solutions provides more options for users, which is crucial in practical applications. Hence, using TPR and TNR as two objective functions, the designed NSGA-II method can obtain satisfactory Pareto fronts. This indicates that our designed encoding method, crossover operation, and mutation operation can effectively handle the established mixed-variable multi-objective optimization problem.

The combination scheme of clustering strategies obtained for SWaT and WADI in the backdoor attack scenarios of SIG and BadNet are as follows: For SWaT: [(1, 0.3), (10, 0.3),

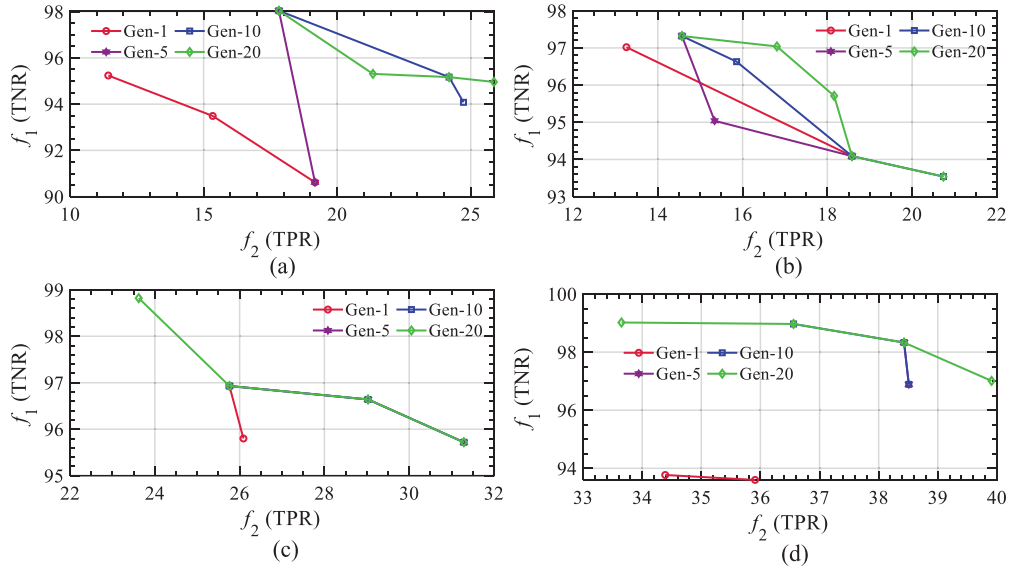


Fig. 8. The Pareto front of the proposed MoCC-BD-FID method. (a) SWaT dataset under SIG attack; (b) SWaT dataset under BadNet attack; (c) WADI dataset under SIG attack; (d) WADI dataset under BadNet attack. Gen- i , $i = 1, 5, 10, 20$ represents the i th generation.

(12, 0.4)] and [(11, 0.4), (12, 0.6)]. For WADI: [(3, 0.1), (6, 0.5), (12, 0.4)] and [(1, 0.2), (11, 0.3), (12, 0.5)]. Fig. 8 shows the corresponding combination scheme of clustering strategies. For example, in the SIG attack scenario of the SWaT dataset, the combination scheme of clustering strategies is [(1, 0.3), (10, 0.3), (12, 0.4)], which indicates the use of three combination strategies: HDBSCAN, Agglomerative Clustering, and Affinity Propagation and the corresponding confidence levels are 0.3, 0.3, and 0.4.

D. Comparison With Single Clustering-Based Defense

First, collect sample parameters of normal and malicious updates generated during the SIG and BadNet backdoor attacks to serve as a training set for offline optimization in the defense phase. To enhance the robustness of the backdoor defense, the training set includes samples that each contain 10% of malicious update parameters produced by SIG and BadNet. The optimized combination scheme of clustering strategies is then applied to two different scenarios, i.e., SIG and BadNet.

1) *Backdoor Defense for SWaT Dataset*: Table IV and Table V present the backdoor defense experimental results on the SWaT dataset. It should be noted that the minor differences in MA metrics in Table VI are related to our designed backdoor attack approach. A successful backdoor attack should maintain model performance on normal samples while effectively triggering specific backdoor samples. This means metrics like accuracy and F1-score on normal test sets should show no significant difference from clean models, while accurately identifying backdoor samples.

The BA primarily depends on the clustering methods employed. The experiments reveal that single clustering methods can initially identify normal gradients, i.e., high TNR, precisely in early federated training rounds due to their dominant proportion. However, if any round exhibits poor clustering performance on normal gradients, the aggregated

TABLE IV
COMPARISON OF BACKDOOR DEFENSE BY DIFFERENT METHODS AGAINST SIG BACKDOOR ATTACK ON THE SWaT DATASET

Method	MA	BA	TPR	TNR
No-defense	0.9883	0.5284	0.0000	0.9000
HDBSCAN	0.9882	0.5033	0.0879	0.9108
DBSCAN	0.9879	0.4793	0.0000	0.9000
K-Means	0.9864	0.501	0.1043	0.8964
MeanShift	0.9679	0.4212	0.1168	0.912
OPTICS	0.9865	0.4832	0.0799	0.8956
SpectralClustering	0.9884	0.3985	0.0894	0.9019
GaussianMixture	0.9835	0.4195	0.0600	0.8755
MiniBatchKMeans	0.9849	0.4852	0.0986	0.905
Birch	0.9847	0.4021	0.1022	0.8974
AgglomerativeClustering	0.9819	0.3782	0.1203	0.9008
AffinityPropagation	0.9843	0.3875	0.1032	0.9106
SOS	0.9881	0.3028	0.1173	0.9239
MoCC-BD-FID	0.9883	0.2664	0.1200	0.9357

TABLE V
COMPARISON OF BACKDOOR DEFENSE BY DIFFERENT METHODS AGAINST BADNET BACKDOOR ATTACK ON THE SWaT DATASET

Method	MA	BA	TPR	TNR
No-defense	0.9884	0.6136	0.000	0.9000
HDBSCAN	0.9884	0.5211	0.1173	0.9173
DBSCAN	0.9894	0.5963	0.1011	0.9090
K-Means	0.9869	0.5248	0.0509	0.8850
MeanShift	0.9867	0.5422	0.0000	0.9000
OPTICS	0.9835	0.3670	0.0982	0.9413
SpectralClustering	0.9863	0.5165	0.0401	0.8800
GaussianMixture	0.9846	0.5505	0.0365	0.8751
MiniBatchKMeans	0.9859	0.5101	0.0723	0.8901
Birch	0.9874	0.5431	0.0863	0.8958
AgglomerativeClustering	0.9882	0.4633	0.1155	0.9080
AffinityPropagation	0.9862	0.4294	0.0989	0.8920
SOS	0.9905	0.4202	0.1173	0.9173
MoCC-BD-FID	0.9882	0.2185	0.1158	0.9586

global model incorporates more backdoor parameters, subsequently increasing difficulty in identifying normal gradients in

TABLE VI

COMPARISON OF BACKDOOR DEFENSE BY DIFFERENT METHODS AGAINST SIG BACKDOOR ATTACK ON THE WADI DATASET

Method	MA	BA	TPR	TNR
No-defense	0.9973	0.2200	0.0000	0.9000
HDBSCAN	0.9979	0.1900	0.0558	0.8930
DBSCAN	0.9956	0.4100	0.0321	0.9100
K-Means	0.9976	0.1800	0.1252	0.9081
MeanShift	0.9977	0.2200	0.1732	0.9237
OPTICS	0.9964	0.7200	0.1067	0.8834
SpectralClustering	0.9978	0.2200	0.0714	0.8889
GaussianMixture	0.9976	0.2800	0.0462	0.8848
MiniBatchKMeans	0.9970	0.2000	0.1729	0.9219
Birch	0.9975	0.2300	0.1248	0.9078
AgglomerativeClustering	0.9976	0.2200	0.0000	0.8795
AffinityPropagation	0.9971	0.1200	0.1093	0.9545
SOS	0.9975	0.0800	0.1380	0.9380
MoCC-BD-FID	0.9978	0.0300	0.1483	0.9676

later rounds. This leads to gradually deteriorating clustering performance, demonstrating the limitation of single clustering methods, i.e., their lack of robustness. A single round of poor clustering can significantly compromise overall defense effectiveness, causing BA to increase and stabilize at higher values while TNR approaches undefended states. This explains why most single clustering methods show similar BA and TNR metrics. In contrast, the proposed combined multiple clustering approach enhances robustness, where poor performance of any single method in a particular round doesn't affect overall clustering effectiveness, thereby improving TNR and reducing BA.

Regarding TPR metrics, the generally low values in the tables result from our experimental setup where malicious clients constituted only 10% of SWaT dataset participants, with backdoor samples accounting for merely 0.5% of each malicious client's training data. This configuration makes backdoor attacks stealthier, causing most clustering methods to perform poorly in identifying malicious gradients, hence the similarly low TPR values across different methods.

From Tables IV–V, it can be observed that:

MoCC-BD-FID achieves the best overall performance. Although the TPR value of Agglomerative Clustering is 0.1203, slightly higher than the 0.12 of MoCC-BD-FID, all other single cluster-based defense methods performed worse than MoCC-BD-FID. Specifically, when comparing MoCC-BD-FID with Spectral Clustering, it can be observed that all other performance indices of MoCC-BD-FID are much better than those of Spectral Clustering. Similarly, when comparing MoCC-BD-FID with Agglomerative Clustering, the other performance indices of MoCC-BD-FID are significantly better. This shows that the proposed MoCC-BD-FID has a strong defense effect against SIG backdoor attacks in the SWaT dataset.

Comparing the defense methods based on single cluster strategy with the no-defense scenario, the BA value under the no-defense condition is the worst. Intuitively, since no-defense does not account for defending against backdoor attacks, its BA value is the highest. The results obtained by MoCC-BD-FID are based on a combination of three clustering

strategies with different confidence levels. Compared with a single clustering strategy, MoCC-BD-FID achieves the lowest BA value, as it can better defend against backdoor attacks. Moreover, these three clustering strategies are automatically selected without requiring manual trial and error, which is one of the advantages of MoCC-BD-FID.

By analyzing the results of single cluster strategy, it can be seen that no single cluster strategy method can achieve the best result. This also indicates the necessity of selecting an appropriate combination scheme of clustering strategies.

In the case of the BadNet backdoor attack, we can also obtain similar results. The MoCC-BD-FID method achieves the best results. Although its TPR value is not as good as HDBSCAN and SOS, this performance index of MoCC-BD-FID are very close to those of HDBSCAN and SOS. Further comparison between MoCC-BD-FID and SOS shows that the BA value of MoCC-BD-FID is 0.2185, much smaller than the BA value of SOS. Additionally, the TNR value of MoCC-BD-FID is 0.9586, which is much higher than the 0.9173 of SOS.

Comparing the defense performances in the SIG and BadNet scenarios, we can see that the BA values in the SIG scenario are all larger than those in the BadNet scenario, indicating that the SIG attack has a more severe impact. However, MoCC-BD-FID can still maintain the BA value below 0.3, demonstrating the effectiveness of MoCC-BD-FID in defending against backdoor attacks.

Regarding the issue of low TPR and TNR accuracy in single cluster strategy-based backdoor defense method, the proposed MoCC-BD-FID improves performance and surpasses all single cluster strategy-based methods, except that under SIG attacks, the TPR is lower than DBSCAN and Spectral Clustering. Here, we need to emphasize that although we consider various clustering strategies, the combination scheme is automatically selected and does not require trial-and-error method.

2) *Backdoor Defense for WADI Dataset:* Tables VI shows the defense results on the WADI dataset under SIG backdoor attack. From Table VI, we can see that MoCC-BD-FID achieves the best overall performance. Although the TPR value is 0.1483, ranking third, it is only slightly worse than MeanShift and MiniBatchKMeans but better than other strategies. Analyzing the BA value, we observe that MoCC-BD-FID's value of 0.03 is significantly better than other methods, especially compared to OPTICS, which has a BA value as high as 0.72. Furthermore, the results reveal that MeanShift performs best in TPR. SOS achieves the best BA result among single clustering strategies, but its TPR and TNR are not the best. This shows that single clustering strategies may perform well in certain aspects but struggle to achieve the best overall performance. MoCC-BD-FID, targeting the SIG backdoor attack on the WADI dataset, automatically selects the combination scheme of clustering strategies with corresponding confidence levels, achieving better overall performance than single clustering strategies while overcoming the design process relied on experts.

Regarding the scenario of BadNet backdoor attacks on the WADI dataset, Table VII presents the comparison results of MoCC-BD-FID and other single clustering strategies. From

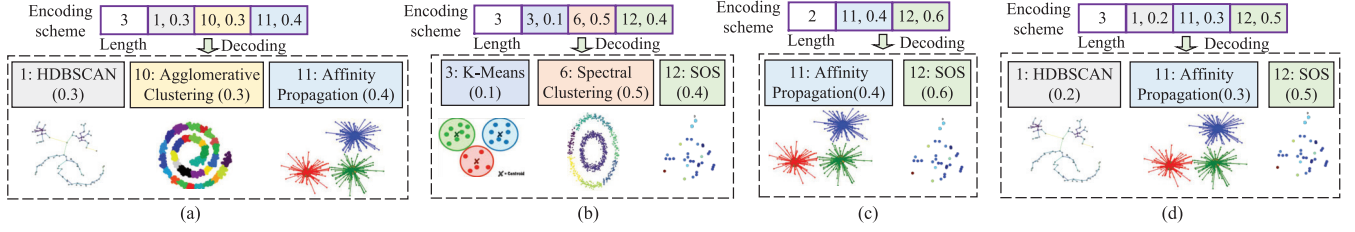


Fig. 9. The final clustering combination strategy obtained from Pareto front of the proposed MoCC-BD-FID method. (a) SWaT dataset under SIG attack; (b) SWaT dataset under BadNet attack; (c) WADI dataset under SIG attack; (d) WADI dataset under BadNet attack.

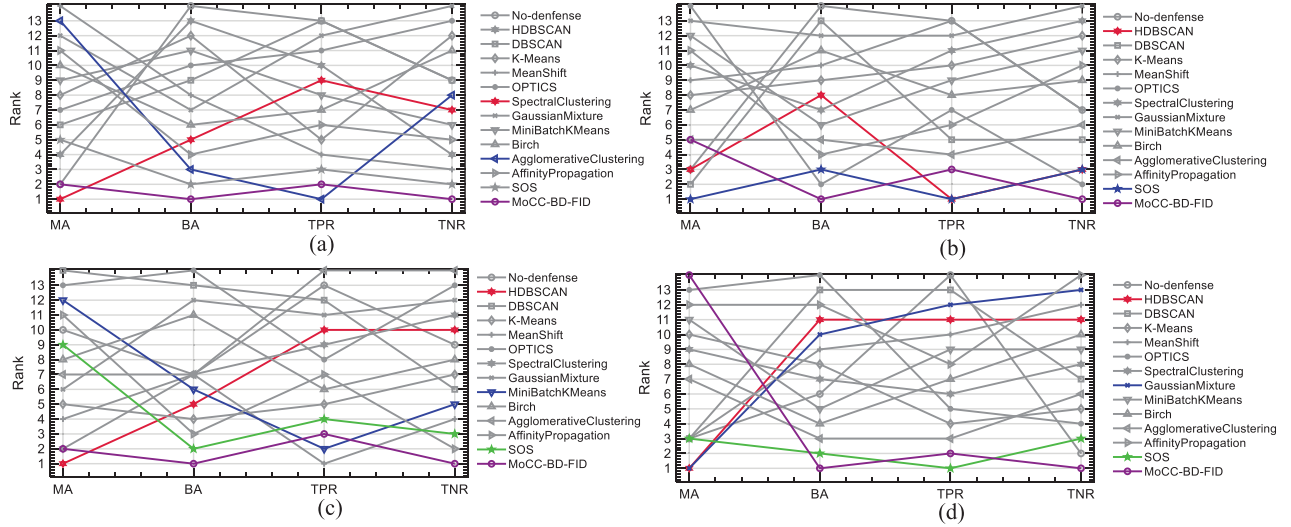


Fig. 10. The rank of four performance indices of MoCC-BD-FID and other compared methods: (a) SIG backdoor attack on SWaT dataset; (b) BadNet backdoor attack on SWaT dataset; (c) SIG backdoor attack on WADI dataset; (d) BadNet backdoor attack on WADI dataset.

TABLE VII

COMPARISON OF BACKDOOR DEFENSE BY DIFFERENT METHODS AGAINST BADNET BACKDOOR ATTACK ON THE WADI DATASET

Method	MA	BA	TPR	TNR
No-denfense	0.9978	0.5075	0.0000	0.9000
HDBSCAN	0.9979	0.5900	0.1343	0.7914
DBSCAN	0.9978	0.7925	0.0282	0.808
K-Means	0.9972	0.5375	0.2246	0.8191
MeanShift	0.9978	0.5425	0.1548	0.784
OPTICS	0.9970	0.8200	0.2021	0.8216
SpectralClustering	0.9973	0.5250	0.1973	0.7995
GaussianMixture	0.9979	0.5875	0.1244	0.7760
MiniBatchKMeans	0.9972	0.4825	0.1887	0.799
Birch	0.9974	0.3725	0.1917	0.7971
AgglomerativeClustering	0.9977	0.3025	0.2400	0.8133
AffinityPropagation	0.9971	0.7375	0.1913	0.7468
SOS	0.9978	0.2025	0.2900	0.8900
MoCC-BD-FID	0.9882	0.1682	0.2577	0.9187

Table VII, it can be seen that MoCC-BD-FID has the best overall performance. Especially, in the BA value, it achieves as low as 0.1682, which is significantly better than the highest value of 0.7925. The TPR metric ranks second, just behind SOS's 0.29. Additionally, MoCC-BD-FID achieves the best TNR, indicating that MoCC-BD-FID is effective in handling BadNet attacks.

Fig. 10 gives the rank of four performance indices of MoCC-BD-FID and other compared method. From Fig. 10, it can be seen that most performance indices of the

TABLE VIII

COMPARISON OF FIVE EXISTING BACKDOOR DEFENSES AGAINST SIG BACKDOOR ATTACK ON THE SWaT DATASET

Method	MA	BA	TPR	TNR
Krum [34]	0.9833	0.4591	-	-
Weak-DP [35]	0.9791	0.4108	-	-
FoolsGold [36]	0.9856	0.4685	-	-
DeepSight [37]	0.9893	0.3972	0.1075	0.9081
CrowdGuard [22]	0.9884	0.4986	0.1193	0.9043
MoCC-BD-FID	0.9883	0.2664	0.1200	0.9357

MoCC-BD-FID method rank first or second, especially the BA, TPR, and TNR metrics. Since MoCC-BD-FID is essentially a backdoor defense method, the BA metric is very important and always ranks first. The TPR and TNR metrics, which are the objective functions during evolutionary process, can dominate most methods. This indicates the effectiveness of MoCC-BD-FID.

E. Comparison With Existing Backdoor Defense

To further demonstrate the effectiveness of the proposed backdoor defense method, we select five existing backdoor defense methods including three classic backdoor defense methods, i.e., Krum [34], Weak-DP [35], FoolsGold [36] and two manually designed methods, i.e., DeepSight [37] and CrowdGuard [22], as comparison methods on the SWaT and WADI datasets.

TABLE IX

COMPARISON OF FIVE EXISTING BACKDOOR DEFENSES AGAINST BAD-NET BACKDOOR ATTACK ON THE SWAT DATASET

Method	MA	BA	TPR	TNR
Krum [34]	0.9885	0.4639	-	-
Weak-DP [35]	0.9735	0.4074	-	-
FoolsGold [36]	0.9881	0.5422	-	-
DeepSight [37]	0.9872	0.5872	0.1025	0.9348
CrowdGuard [22]	0.988	0.423	0.1048	0.9031
MoCC-BD-FID	0.9882	0.2185	0.1158	0.9586

TABLE X

COMPARISON OF FIVE EXISTING BACKDOOR DEFENSES AGAINST SIG BACKDOOR ATTACK ON THE WADI DATASET

Method	MA	BA	TPR	TNR
Krum [34]	0.9969	0.2200	-	-
Weak-DP [35]	0.9773	0.1600	-	-
FoolsGold [36]	0.9972	0.2000	-	-
DeepSight [37]	0.9966	0.1200	0.1272	0.923
CrowdGuard [22]	0.9974	0.0800	0.1408	0.9045
MoCC-BD-FID	0.9978	0.0300	0.1483	0.9676

TABLE XI

COMPARISON OF FIVE EXISTING BACKDOOR DEFENSES AGAINST BAD-NET BACKDOOR ATTACK ON THE WADI DATASET

Method	MA	BA	TPR	TNR
Krum [34]	0.9973	0.4429	-	-
Weak-DP [35]	0.9975	0.3805	-	-
FoolsGold [36]	0.9967	0.632	-	-
DeepSight [37]	0.9973	0.4391	0.2062	0.8812
CrowdGuard [22]	0.9971	0.3994	0.1851	0.8503
MoCC-BD-FID	0.9882	0.1682	0.2577	0.9187

Table VIII and Table IX show the comparison of six defense methods under SIG backdoor attack and BadNet backdoor attack on the SWaT dataset, respectively. From Table VIII, MoCC-BD-FID outperforms the existing five methods. Although the MA value of MoCC-BD-FID is 0.9883, slightly lower than DeepSight's 0.9893 and CrowdGuard's 0.9884, the other metrics are all the best. It should be noted that the BA metric of MoCC-BD-FID is significantly higher than that of manually designed CrowdGuard and DeepSight. This shows that the method of automatically selecting a combination scheme of clustering strategies can effectively defend against SIG backdoor attacks. Similarly, from Table IX, we can see that MoCC-BD-FID is only slightly lower than Krum's 0.9885 in the MA metric, while it performs best in all other aspects, especially with a significant improvement in the BA value.

Table X and Table XI present the defense comparison results of MoCC-BD-FID and five other methods under SIG and BadNet attacks on the WADI dataset, respectively. From Table X, it can be observed that all the metrics of MoCC-BD-FID are the best, with significant improvements in both the BA and TNR metrics. From Table XI, although the MA value of MoCC-BD-FID is slightly inferior to other methods, its BA, TPR, and TNR values are much better than those of the other methods.

TABLE XII

COMPARISON OF BACKDOOR DEFENSE BY DIFFERENT METHODS AGAINST SIG BACKDOOR ATTACK ON THE SPA DATASET

Method	MA	BA	TPR	TNR
No-defense	0.9577	0.4132	0.0000	0.8000
HDBSCAN	0.9563	0.4397	0.0736	0.6773
DBSCAN	0.9571	0.4556	0.0000	0.8000
K-Means	0.9532	0.3841	0.1750	0.7839
MeanShift	0.9534	0.4238	0.0000	0.8000
OPTICS	0.9544	0.4794	0.0000	0.8000
SpectralClustering	0.9540	0.5902	0.0600	0.7410
GaussianMixture	0.9569	0.5429	0.1823	0.7810
MiniBatchKMeans	0.9582	0.5111	0.1016	0.7930
Birch	0.9571	0.4317	0.0557	0.7256
AgglomerativeClustering	0.9521	0.5746	0.0500	0.7296
AffinityPropagation	0.9577	0.2968	0.1934	0.7976
SOS	0.9528	0.4714	0.0000	0.8000
MoCC-BD-FID	0.9569	0.2062	0.1942	0.8409

TABLE XIII

COMPARISON OF BACKDOOR DEFENSE BY DIFFERENT METHODS AGAINST BADNET BACKDOOR ATTACK ON THE SPA DATASET

Method	MA	BA	TPR	TNR
No-defense	0.9584	0.3297	0.0000	0.8000
HDBSCAN	0.9559	0.3627	0.0000	0.6813
DBSCAN	0.9585	0.3371	0.0000	0.8000
K-Means	0.9573	0.3606	0.1400	0.7514
MeanShift	0.9581	0.3154	0.0000	0.8000
OPTICS	0.9569	0.3351	0.0000	0.8000
SpectralClustering	0.9555	0.3468	0.1600	0.7436
GaussianMixture	0.9542	0.3571	0.0000	0.6584
MiniBatchKMeans	0.9536	0.3568	0.0180	0.7244
Birch	0.9586	0.2216	0.2366	0.8184
AgglomerativeClustering	0.9561	0.3453	0.0500	0.7339
AffinityPropagation	0.9567	0.3586	0.0713	0.6353
SOS	0.9571	0.3371	0.0000	0.8000
MoCC-BD-FID	0.9584	0.1908	0.2164	0.8519

F. Transferability

This subsection demonstrates the transferability of MoCC-BD-FID. While previous experimental results focused on Singapore's ICSs, we here employ the PSA dataset from Oak Ridge National Laboratories in the USA. Tables XII and XIII present the backdoor defense results against BadNet backdoor attack and SIG backdoor attack on the PSA dataset. The results show that MoCC-BD-FID achieves similar MA values while obtaining the best BA values compared to no-defense and other single clustering methods. For TPR and TNR metrics, MoCC-BD-FID delivers the best performance against SIG backdoor attack. Against BadNet backdoor attack, MoCC-BD-FID achieves 0.2164 in TPR, slightly lower than the top-performing Birch single clustering method, but significantly outperforms other single clustering approaches.

To further demonstrate the effectiveness of MoCC-BD-FID on the PSA dataset, we compared it with Krum [34], Weak-DP [35], FoolsGold [36], DeepSight [37], and CrowdGuard [22]. Table XIV presents the comparison results between MoCC-BD-FID and these five methods against SIG backdoor attacks on the SPA dataset, while Table XV shows the defense performance comparison of all six methods against BadNet backdoor attacks on SPA dataset. The results in Tables XIV and XV

TABLE XIV

COMPARISON OF FIVE EXISTING BACKDOOR DEFENSES AGAINST SIG BACKDOOR ATTACK ON THE SPA DATASET

Method	MA	BA	TPR	TNR
Krum [34]	0.9551	0.3988	-	-
Weak-DP [35]	0.9577	0.3842	-	-
FoolsGold [36]	0.9573	0.4265	-	-
DeepSight [37]	0.9573	0.3806	0.1208	0.8179
CrowdGuard [22]	0.9567	0.4015	0.1317	0.8024
MoCC-BD-FID	0.9569	0.2062	0.1942	0.8409

TABLE XV

COMPARISON OF FIVE EXISTING BACKDOOR DEFENSES AGAINST SIG BACKDOOR ATTACK ON THE SPA DATASET

Method	MA	BA	TPR	TNR
Krum [34]	0.9579	0.3394	-	-
Weak-DP [35]	0.9581	0.3225	-	-
FoolsGold [36]	0.9559	0.3671	-	-
DeepSight [37]	0.9565	0.3058	0.1143	0.8203
CrowdGuard [22]	0.9573	0.3129	0.1295	0.8017
MoCC-BD-FID	0.9584	0.1908	0.2164	0.8519

demonstrate that MoCC-BD-FID achieves best performance, particularly in terms of the BA metric, obtaining values of 0.2062 and 0.1908 for SIG backdoor attack and BadNet backdoor attack scenarios, respectively. These comprehensive results confirm that MoCC-BD-FID can be successfully transferred to other ICS environments while maintaining excellent performance.

G. Discussion

In real ICSs, online testing of new IDSs is typically not permitted. Performance evaluation of novel IDSs is generally first conducted on public datasets or experimental testbeds. Considering data privacy and security concerns when actual systems are under attack, complete datasets are rarely publicly available. In academic research, experimental approaches can simulate attacks on practical systems to generate highly representative data including the characteristics of large-scale and operational environments [43]. In this paper, we employ the SWaT, WADI, and SPA datasets, which largely reflect the characteristics of real systems, and are widely used in existing literature to validate the effectiveness of model. Therefore, if MoCC-BD-FID performs well on these datasets, it should also achieve good performance when applied to practical systems. In practical deployment, the MoCC-BD-FID is conducted offline. During online detection, the optimized clustering scheme is directly applied without introducing additional latency, thus satisfying real-time response requirements. Furthermore, since FL model aggregation is inherently periodic, the MoCC-BD-FID only needs to be executed once during the initial federated training phase. Subsequent model update cycles do not require multiple iterations. Therefore, when deployed in practical systems, MoCC-BD-FID can achieve real-time detection.

Detecting out of distribution (OOD) backdoors remains a challenging task, and MoCC-BD-FID can be extended for OOD backdoor detection. Following [44], we first inject indicator tasks into the global model by using OOD data,

then leverage the distributional similarity between backdoor samples and OOD samples to obtain the required backdoor samples for MoCC-BD-FID. This enables MoCC-BD-FID to detect OOD backdoors to some extent. Also, MoCC-BD-FID can be extended to solve OOD backdoors by designing a method to mount clean-label backdoor attacks or predicting label transitions based on counterfactual explanation [44], [45].

V. CONCLUSION

In this paper, we have proposed a multi-objective clustering combination-based backdoor defense for federated intrusion detection of ICSs, called MoCC-BD-FID. In MoCC-BD-FID, we establish the automated design issue of clustering strategies combination for defense method as the mixed-variable multi-objective optimization problem by maximizing both TPR and TNR. Besides, we have developed an efficient encoding strategy, crossover operation and mutation operation under the framework of NSGA-II to describe and evolve the different combination schemes of 12 classic clustering strategies with different confidence levels. The effectiveness of the backdoor defense strategy is illustrated by subjecting 1D-CNN to two different attack scenarios including SIG and BadNet on three widely-used ICS intrusion datasets i.e., SWaT, WADI, and PSA. The experimental results have shown that MoCC-BD-FID achieves better performance in MA, BA, TPR and TNR than other 12 single cluster strategy-based methods, three classic backdoor defense methods, i.e., Krum [34], Weak-DP [35], FoolsGold [36] and two manually designed methods, i.e., DeepSight [37] and CrowdGuard [22]. In addition, the combination scheme of clustering strategies is automatically designed by the developed NSGA-II, which overcomes the difficulty in determining combination scheme involving more than two clustering strategies.

This work focuses on the automatic selection of an appropriate combination of clustering strategies for defending against backdoor attacks, so the illustrated deep learning model employed in this study is a simple 1D-CNN, whose detection performance especially the FPR can be further improved. In practical deployments, more advanced models such as automated federated IDS (Fed-GA-CNN-IDS) [14] and multi-objective discrete extremal optimization-based CNN (MODEO-CNN) [47] will be adopted to further reduce the FPR. For federated IDSs, FPR is directly correlated with model performance after federated training. However, FL encounters increasing convergence challenges with more participating clients, making it difficult to obtain optimal models. For future work, we plan to implement multi-dimensional collaborative optimization through personalized FL, efficient aggregation strategies, and automated deep learning techniques to systematically reduce FPR while improving convergence efficiency. Also, it will be interesting to develop more robust defense methods against OOD backdoor attacks.

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